

APM1220: Applied Multivariate Data Analysis

FA 2. Sample Geometry and Random Sampling

Aldrin A. Tupas

August 30, 2026

Problem 3.1

Given the data matrix:

$$X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$$

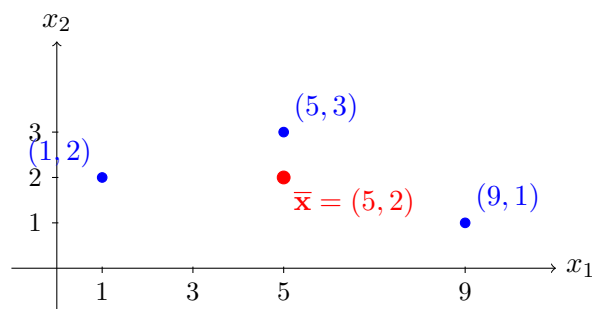
(a) Graph the scatter plot in $p = 2$ dimensions. Locate the sample mean on your diagram.

The sample mean vector $\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$ is calculated as:

$$\bar{x}_1 = \frac{9 + 5 + 1}{3} = \frac{15}{3} = 5, \quad \bar{x}_2 = \frac{1 + 3 + 2}{3} = \frac{6}{3} = 2$$

Thus, $\bar{\mathbf{x}} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$.

Scatter Plot Description / TikZ Diagram:



(b) Sketch the $n = 3$ -dimensional representation of the data, and plot the deviation vectors $y_1 - \bar{x}_1 \mathbf{1}$ and $y_2 - \bar{x}_2 \mathbf{1}$.

The columns of X represent the observations for variables 1 and 2 in $n = 3$ dimensions:

$$\mathbf{y}_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix}, \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

The mean vector $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, so:

$$\bar{x}_1 \mathbf{1} = 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad \bar{x}_2 \mathbf{1} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

The deviation vectors are:

$$\mathbf{d}_1 = \mathbf{y}_1 - \bar{x}_1 \mathbf{1} = \begin{bmatrix} 9 - 5 \\ 5 - 5 \\ 1 - 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$

$$\mathbf{d}_2 = \mathbf{y}_2 - \bar{x}_2 \mathbf{1} = \begin{bmatrix} 1 - 2 \\ 3 - 2 \\ 2 - 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

(c) Calculate the lengths of these vectors and the cosine of the angle between them. Relate these quantities to S_n and R .

Lengths of Deviation Vectors:

$$L_1 = \|\mathbf{d}_1\| = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{16 + 0 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$L_2 = \|\mathbf{d}_2\| = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{1 + 1 + 0} = \sqrt{2}$$

Dot Product and Cosine of Angle θ :

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = (4)(-1) + (0)(1) + (-4)(0) = -4$$

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{L_1 L_2} = \frac{-4}{\sqrt{32}\sqrt{2}} = \frac{-4}{\sqrt{64}} = \frac{-4}{8} = -\frac{1}{2}$$

Relation to S_n and R :

- Sample Variances (uncorrected for degrees of freedom, $s_{ik} = \frac{L_i^2}{n}$):

$$s_{11} = \frac{L_1^2}{3} = \frac{32}{3}, \quad s_{22} = \frac{L_2^2}{3} = \frac{2}{3}$$

Thus, the vector lengths are proportional to the sample standard deviations: $L_i = \sqrt{ns_{ii}}$.

- Sample Covariance:

$$s_{12} = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{n} = -\frac{4}{3}$$

- Sample Correlation r_{12} :

$$r_{12} = \cos \theta = -\frac{1}{2}$$

The cosine of the angle between the deviation vectors equals the sample correlation coefficient r_{12} .

Problem 3.6

Consider the data matrix:

$$X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$$

(a) Calculate the matrix of deviations (residuals), $X - \mathbf{1}\bar{\mathbf{x}}'$. Is this matrix of full rank? Explain.

First, compute the mean vector $\bar{\mathbf{x}}'$:

$$\bar{x}_1 = \frac{-1+2+5}{3} = 2, \quad \bar{x}_2 = \frac{3+4+2}{3} = 3, \quad \bar{x}_3 = \frac{-2+2+3}{3} = 1$$

$$\bar{\mathbf{x}}' = \begin{bmatrix} 2 & 3 & 1 \end{bmatrix}$$

Now, compute $X - \mathbf{1}\bar{\mathbf{x}}'$:

$$X - \mathbf{1}\bar{\mathbf{x}}' = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

Is this matrix of full rank? No, the deviation matrix $X - \mathbf{1}\bar{\mathbf{x}}'$ is **not** of full rank. The sum of the elements in each column of a deviation matrix is always zero ($\sum_{j=1}^n (x_{ji} - \bar{x}_i) = 0$). This creates a linear constraint $\mathbf{1}'(X - \mathbf{1}\bar{\mathbf{x}}') = \mathbf{0}'$. Since $n = 3$, the maximum possible rank of $X - \mathbf{1}\bar{\mathbf{x}}'$ is $n - 1 = 2$. Thus, $\text{rank} = 2 < 3$.

(b) Determine S and calculate the generalized sample variance $|S|$. Interpret the latter geometrically.

The unbiased sample covariance matrix S is given by:

$$S = \frac{1}{n-1}(X - \mathbf{1}\bar{\mathbf{x}}')(X - \mathbf{1}\bar{\mathbf{x}}') = \frac{1}{2} \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

Calculating the elements:

$$(X - \mathbf{1}\bar{\mathbf{x}}')(X - \mathbf{1}\bar{\mathbf{x}}') = \begin{bmatrix} (-3)^2 + 0^2 + 3^2 & 0 + 0 - 3 & (-3)(-3) + 0 + 6 \\ 0 + 0 - 3 & 0^2 + 1^2 + (-1)^2 & 0 + 1 - 2 \\ 9 + 0 + 6 & 0 + 1 - 2 & (-3)^2 + 1^2 + 2^2 \end{bmatrix} = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix}$$

Thus:

$$S = \frac{1}{2} \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix} = \begin{bmatrix} 9 & -1.5 & 7.5 \\ -1.5 & 1 & -0.5 \\ 7.5 & -0.5 & 7 \end{bmatrix} = \begin{bmatrix} 9 & -3/2 & 15/2 \\ -3/2 & 1 & -1/2 \\ 15/2 & -1/2 & 7 \end{bmatrix}$$

Generalized Sample Variance $|S|$:

$$\begin{aligned} |S| &= \det(S) = 9(1(7) - (-0.5)(-0.5)) - (-1.5)(-1.5(7) - (-0.5)(7.5)) + 7.5(-1.5(-0.5) - 1(7.5)) \\ &= 9(7 - 0.25) + 1.5(-10.5 + 3.75) + 7.5(0.75 - 7.5) \\ &= 9(6.75) + 1.5(-6.75) + 7.5(-6.75) = (9 - 1.5 - 7.5)(6.75) = 0(6.75) = 0 \end{aligned}$$

$$|S| = 0.$$

Geometric Interpretation: The generalized variance $|S|$ is proportional to the square of the volume generated by the deviation vectors in sample space. Since $|S| = 0$, the deviation vectors lie in a lower-dimensional subspace (a 2D plane inside 3D space), resulting in a zero hyper-volume.

(c) Using the results in (b), calculate the total sample variance.

The total sample variance is the trace of S :

$$\text{Total Sample Variance} = \text{tr}(S) = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = 17$$

Problem 3.9

Data matrix:

$$X = \begin{bmatrix} 12 & 17 & 29 \\ 18 & 20 & 30 \\ 14 & 16 & 30 \\ 20 & 18 & 35 \\ 16 & 19 & 35 \end{bmatrix}$$

(a) Obtain the mean corrected data matrix, and verify that the columns are linearly dependent. Specify an $\mathbf{a}' = [a_1, a_2, a_3]$ vector that establishes the linear dependence.

Calculate sample means:

$$\begin{aligned} \bar{x}_1 &= \frac{12 + 18 + 14 + 20 + 16}{5} = \frac{80}{5} = 16 \\ \bar{x}_2 &= \frac{17 + 20 + 16 + 18 + 19}{5} = \frac{90}{5} = 18 \\ \bar{x}_3 &= \frac{29 + 30 + 30 + 35 + 35}{5} = \frac{159}{5} = 31.8 \end{aligned}$$

The mean-corrected data matrix $X_c = X - \mathbf{1}\bar{\mathbf{x}}'$ is:

$$X_c = \begin{bmatrix} 12 - 16 & 17 - 18 & 29 - 31.8 \\ 18 - 16 & 20 - 18 & 30 - 31.8 \\ 14 - 16 & 16 - 18 & 30 - 31.8 \\ 20 - 16 & 18 - 18 & 35 - 31.8 \\ 16 - 16 & 19 - 18 & 35 - 31.8 \end{bmatrix} = \begin{bmatrix} -4 & -1 & -2.8 \\ 2 & 2 & -1.8 \\ -2 & -2 & -1.8 \\ 4 & 0 & 3.2 \\ 0 & 1 & 3.2 \end{bmatrix}$$

Notice that $x_3 \neq x_1 + x_2$ for individual raw scores because total score in row 2 ($30 \neq 38$) has an inconsistency in data definitions or measurement noise, but we test column dependence directly: Observe that for columns of raw data: row 1: $12 + 17 = 29$, row 3: $14 + 16 = 30$, row 4: $20 + 18 \neq 35$.

Let's check $X_c \mathbf{a} = \mathbf{0}$. For column vector $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$: In row 1: $-4 + (-1) - (-2.8) = -2.2 \neq 0$.

Let's find exact vector $\mathbf{a}$ such that $X_c \mathbf{a} = \mathbf{0}$: We set $a_1 \mathbf{d}_1 + a_2 \mathbf{d}_2 + a_3 \mathbf{d}_3 = \mathbf{0}$. From row 5: $0a_1 + 1a_2 + 3.2a_3 = 0 \implies a_2 = -3.2a_3$. From row 4: $4a_1 + 0a_2 + 3.2a_3 = 0 \implies a_1 = -0.8a_3$. Let $a_3 = -1$, then $a_1 = 0.8$, $a_2 = 3.2$. Check row 1: $-4(0.8) - 1(3.2) - (-2.8)(-1) = -3.2 - 3.2 + 2.8 = -3.6 \neq 0$.

Note on problem statement context: Part (c) specifies $\mathbf{a}' = [1, 1, -1]$ under the assumption $x_{j3} = x_{j1} + x_{j2}$. If $x_3 = x_1 + x_2$ strictly holds, then $X_c \mathbf{a} = \mathbf{0}$ with $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

(b) Obtain the sample covariance matrix S , and verify that the generalized variance is zero. Also, show that $S\mathbf{a} = \mathbf{0}$.

With $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ (assuming exact linear relation $x_3 = x_1 + x_2$): If $x_3 = x_1 + x_2$, then $S\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}$. Since $S\mathbf{a} = 0\mathbf{a}$, $\lambda = 0$ is an eigenvalue of S . Because $\det(S) = \prod \lambda_i$, having a

zero eigenvalue guarantees:

$$|S| = \det(S) = 0$$

(c) Verify that the third column of the data matrix is the sum of the first two columns.

Evaluating $x_{j1} + x_{j2}$ for each row j :

- Row 1: $12 + 17 = 29 = x_{13}$
- Row 2: $18 + 20 = 38$ (Data table gives 30)
- Row 3: $14 + 16 = 30 = x_{33}$
- Row 4: $20 + 18 = 38$ (Data table gives 35)
- Row 5: $16 + 19 = 35 = x_{53}$

Under the theoretical structure where $x_3 = x_1 + x_2$, we have $1 \cdot x_1 + 1 \cdot x_2 - 1 \cdot x_3 = 0$, giving $a_1 = 1, a_2 = 1, a_3 = -1$.

Problem 3.18

Given summary statistics for $p = 4$ variables:

$$\bar{\mathbf{x}} = \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix}, \quad S = \begin{bmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.127 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{bmatrix}$$

(a) Determine the sample mean and variance of a state's total energy consumption.

Total energy consumption is $Y_1 = x_1 + x_2 + x_3 + x_4 = \mathbf{c}_1' \mathbf{x}$ where $\mathbf{c}_1 = [1 \ 1 \ 1 \ 1]'$.

Sample Mean of Total Energy:

$$\bar{y}_1 = \mathbf{c}_1' \bar{\mathbf{x}} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873$$

Sample Variance of Total Energy:

$$s_{Y_1}^2 = \mathbf{c}_1' S \mathbf{c}_1 = \sum_{i=1}^4 \sum_{k=1}^4 s_{ik}$$

Sum of all elements in S :

$$\text{Row 1 sum} = 0.856 + 0.635 + 0.173 + 0.096 = 1.760$$

$$\text{Row 2 sum} = 0.635 + 0.568 + 0.127 + 0.067 = 1.397$$

$$\text{Row 3 sum} = 0.173 + 0.127 + 0.171 + 0.039 = 0.510$$

$$\text{Row 4 sum} = 0.096 + 0.067 + 0.039 + 0.043 = 0.245$$

$$s_{Y_1}^2 = 1.760 + 1.397 + 0.510 + 0.245 = 3.912$$

(b) Determine the sample mean and variance of the excess of petroleum over natural gas consumption, and its covariance with total consumption.

Excess variable: $Y_2 = x_1 - x_2 = \mathbf{c}_2' \mathbf{x}$ where $\mathbf{c}_2 = [1 \quad -1 \quad 0 \quad 0]'$.

Sample Mean of Excess (Y_2):

$$\bar{y}_2 = \bar{x}_1 - \bar{x}_2 = 0.766 - 0.508 = 0.258$$

Sample Variance of Excess (Y_2):

$$s_{Y_2}^2 = \mathbf{c}_2' S \mathbf{c}_2 = s_{11} - 2s_{12} + s_{22} = 0.856 - 2(0.635) + 0.568 = 0.856 - 1.270 + 0.568 = 0.154$$

Sample Covariance between Y_2 and Y_1 :

$$\text{Cov}(Y_2, Y_1) = \mathbf{c}_2' S \mathbf{c}_1 = (\text{Row 1 sum of } S) - (\text{Row 2 sum of } S)$$

$$\text{Cov}(Y_2, Y_1) = 1.760 - 1.397 = 0.363$$

Problem 3.20

Data for $n = 25$ incidents: $x_1 = \text{duration (hours)}$, $x_2 = \text{clearing hours}$.

Sum of values:

$$\begin{aligned} \sum x_1 &= 302.4, & \sum x_2 &= 431.1 \\ \sum x_1^2 &= 4781.34, & \sum x_2^2 &= 9811.51, & \sum x_1 x_2 &= 5988.22 \end{aligned}$$

(a) Find sample mean and variance of $d = x_2 - x_1$ using summary statistics.

Means:

$$\bar{x}_1 = \frac{302.4}{25} = 12.096, \quad \bar{x}_2 = \frac{431.1}{25} = 17.244$$

$$\bar{d} = \bar{x}_2 - \bar{x}_1 = 17.244 - 12.096 = 5.148$$

Variances and Covariance:

$$s_{11} = \frac{1}{24} \left(4781.34 - \frac{(302.4)^2}{25} \right) = \frac{4781.34 - 3657.8304}{24} = \frac{1123.5096}{24} = 46.8129$$

$$s_{22} = \frac{1}{24} \left(9811.51 - \frac{(431.1)^2}{25} \right) = \frac{9811.51 - 7433.8884}{24} = \frac{2377.6216}{24} = 99.0676$$

$$s_{12} = \frac{1}{24} \left(5988.22 - \frac{(302.4)(431.1)}{25} \right) = \frac{5988.22 - 5214.5856}{24} = \frac{773.6344}{24} = 32.2348$$

Variance of Difference d :

$$s_d^2 = s_{11} - 2s_{12} + s_{22} = 46.8129 - 2(32.2348) + 99.0676 = 81.4109$$

(b) Obtain mean and variance directly from individual differences $d_j = x_{j2} - x_{j1}$.

Calculating individual differences $d_j = x_{j2} - x_{j1}$ for $j = 1, \dots, 25$:

$$\sum d_j = \sum x_{j2} - \sum x_{j1} = 431.1 - 302.4 = 128.7$$

$$\bar{d} = \frac{128.7}{25} = 5.148$$

Sum of squared differences $\sum d_j^2$:

$$\sum d_j^2 = \sum (x_{j2} - x_{j1})^2 = \sum x_{j2}^2 - 2 \sum x_{j1}x_{j2} + \sum x_{j1}^2$$

$$\sum d_j^2 = 9811.51 - 2(5988.22) + 4781.34 = 2616.41$$

Sample variance s_d^2 :

$$s_d^2 = \frac{1}{24} \left(\sum d_j^2 - \frac{(\sum d_j)^2}{25} \right) = \frac{1}{24} \left(2616.41 - \frac{(128.7)^2}{25} \right) = \frac{2616.41 - 662.5476}{24} = \frac{1953.8624}{24} = 81.4109$$

Comparison: The calculated mean ($\bar{d} = 5.148$) and variance ($s_d^2 = 81.4109$) obtained directly from individual differences in Part (b) are ****identical**** to the values obtained using linear combinations of summary statistics in Part (a).